

## Concept of Abstract Data Types (ADT)

Q. Explain the concept of Abstract Data Type (ADT) with suitable examples. Why is it important in data structures?

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**\*\*Definition to Remember\*\***

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### 1. Key Concepts of ADT

\* **Abstract** = Implementation details are hidden from the user.

\* **Encapsulation**: Data and operations are bundled together.

### 2. How ADT Works

[ Client Program ]

(Calls: push, pop, peek - does NOT know how they work)

ADT Interface      <- Public API (what)

[ Hidden Implementation ]

### 3. Classic Example - Stack ADT

| Component | Detail |

The user calls `push()` and `pop()` without knowing if the stack is internally implemented using an array or a linked list - that detail is abstracted away.

### 4. Advantages of ADT

\* **Code Reusability**: ADT implementation can be reused across different programs.

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***\*\*Must-Write Points (for 10 marks)\*\****

1. ADT

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***\*\*Quick Recall\*\****

`ADT ->  
Reusab